Assignment 2AD, 2025

DD2434 Machine Learning, Advanced Course

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D Section

Question D.1

I chose the dot product similarity as similarity function

$$Y = WX \quad Y \in R^{d \times n}$$

d representes the dimensionality of the data while n is the number of data in the data set and k is the new dimensionality of the data.
We know that W is orthonormal ($W^T W = I$)

$$S = Y^T Y = (WX)^T (WX) = X^T (W^T W) X = X^T X$$

S matrix is the Gram matrix.
Why this choice is meaningful:

- Interpretation →the dot product measures the alignment between two MEPs' voting patterns. Higher values indicate greater agreement.
- Domain appropriateness → in voting behavior analysis, we care about agreement on specific votes. The dot product naturally captures co-voting patterns (when both MEPs vote the same way, it contributes positively to similarity)
- Simplicity

Transformation steps required for our similarity computation:

- Encoding of the EPG column
  - 0 (not an MEP) → exclude from computation
  - 1 (for) → +1
  - 2 (against) → -1

  - 3 (abstention) $\rightarrow$ 0
  - 4 (absent) $\rightarrow$ 0
  - 5 (did not vote) $\rightarrow$ 0
  - 6 (motivated) $\rightarrow$ 0

  "For" and "Against" are opposite positions (+1, -1)

- Filtering $\rightarrow$ MEPs with a very low participation rate are ignored
- Missing Data $\rightarrow$ for each pair of MEPs, only include votes where BOTH MEPs were present and voted. This ensures we're measuring similarity based on actual voting decisions, not absence patterns.
- Center $\rightarrow$ subtract each MEP's mean vote in order to remove the baseline voting tendency of each MEP.
- Normalize $\rightarrow$ divide by L2 norm in order to make similarity values comparable across MEP pairs with different participation rates.

I selected MEPs from both the EP8 (2014-2019) and EP9 (2019-2022) sessions, applying a minimum participation threshold of 30% This resulted in a subset of 1,520 MEPs who voted on 10,251 recorded votes, yielding approximately 2.3 million pairwise similarity comparisons. Reasons:

- combining two consecutive parliamentary sessions provides a broader temporal perspective, allowing the analysis to capture both stable political alignments and potential shifts in voting behavior across time.
- the 30% participation threshold ensures that similarity estimates are based on a substantial number of co-voted issue
- the computed similarity matrix shows expected properties: perfect self-similarity (diagonal ≈ 1.0), symmetry, and a reasonable distribution of similarities (mean = 0.14, std = 0.27) with both high agreement (max = 0.998) and strong opposition (min = -0.42) represented.

Question D.2

We want to find the latent matrix $X \in R^{k\times n}$ such that $S = X^T X$.
The similarity matrix S is symmettic positive semi-definite , so its eigen-decomposition is

$$S = U\Lambda U^{-1} = U\Lambda U^T = (\Lambda^{0.5}U^T)^T(\Lambda^{0.5}U^T)$$

The non negative eigenvalues of S are sorted in descending oder in the diagonal of Λ, we can take the $k \times n$ matrix X to be

$$X = I_{k\times n}\Lambda^{0.5}U^T$$

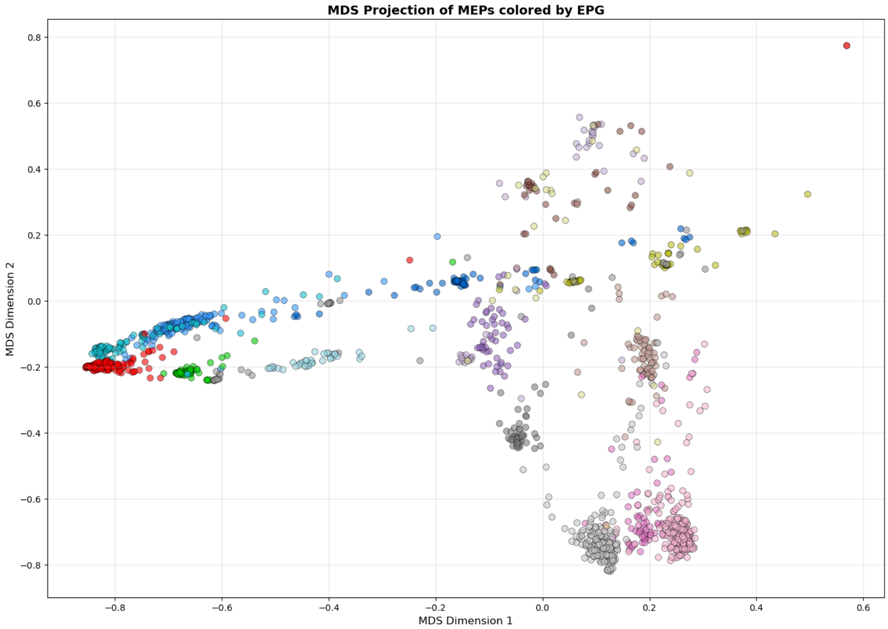

Figure 1: MEPs colored by Political Group (EPG)

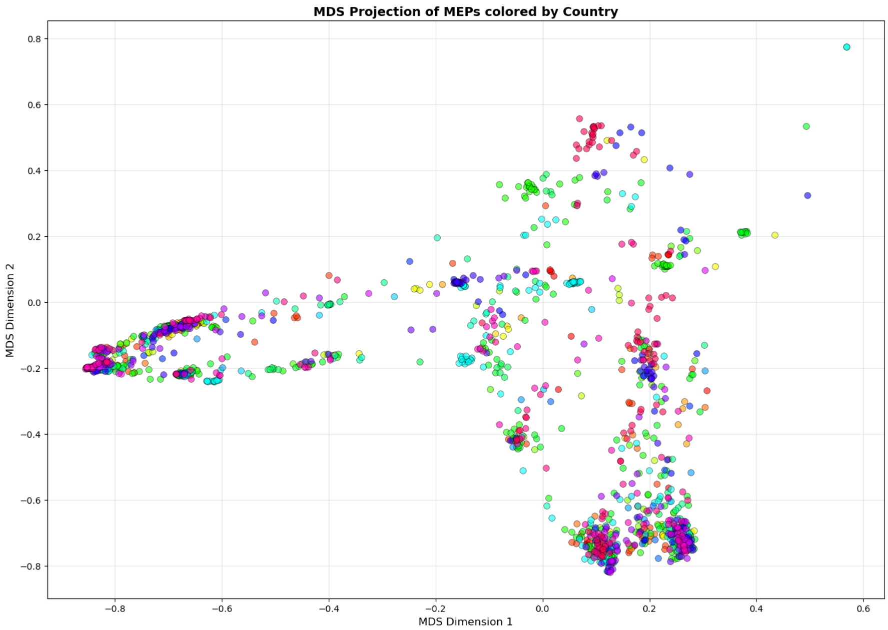

Figure 2: MEPs colored by Country

In my model , I assumed $k = 2$

The MDS visualization reveals that political group (EPG) affiliation is the dominant

factor explaining MEP voting patterns. When colored by EPG, the plot shows clear, well-separated clusters following a classic left-right political spectrum with a distinctive horseshoe pattern. In contrast, when colored by country, the visualization shows extensive overlap with no coherent national clusters.
This finding indicates that MEPs vote primarily along ideological lines rather than national interests. The strong cohesion within European political groups (e.g., EPP, S&D) suggests that party discipline and shared political values transcend national boundaries in the European Parliament.
The horseshoe pattern is particularly noteworthy: the left-center-right spectrum is clearly visible, with centrist parties (EPP, Renew) at the top of the arc, left-wing parties (S&D, Greens, GUE/NGL) on the left side, and right-wing/Eurosceptic parties (ECR, ID) on the right. The fact that political extremes appear closer to each other than to the center likely reflects their shared Eurosceptic positions despite opposite economic ideologies.

Question D.3

In this section, we aim to conduct a more nuanced analysis of the MEP voting dataset by addressing the following research questions:

> *Do Members of the European Parliament (MEPs) tend to vote along party lines when a vote is sponsored by their own political group? Furthermore, which political groups exhibit systematic voting alliances?*

Hypothesis:

- Party loyalty → MEPs will vote overwhelmingly in favor of proposals sponsored by their own political group (high in-group support)
- Coalition patterns → certain political groups will consistently support each other's proposals, revealing stable legislative alliances (e.g., EPP-S&D "grand coalition", or Greens-S&D left-wing cooperation)
- Opposition patterns → ideologically distant groups will systematically vote against each other's proposals (e.g., far-left vs far-right, or Eurosceptics vs federalists)

I used the support matrix with the dataset composed by EP9_RCVs_2022_06_22 and EP9_Voted_docs.
Support matrix is a quantitative tool that measures how political groups vote on each other's legislative proposals. It is constructed as follows:

- Rows → Political groups acting as sponsors (proposing votes/amendments)
- Columns → Political groups acting as voters (voting on those proposals)
- Cell values → support rate computed as $SupportRate = \frac{\text{Number of "For" votes by voter group}}{\text{Total votes cast by voter group}}$

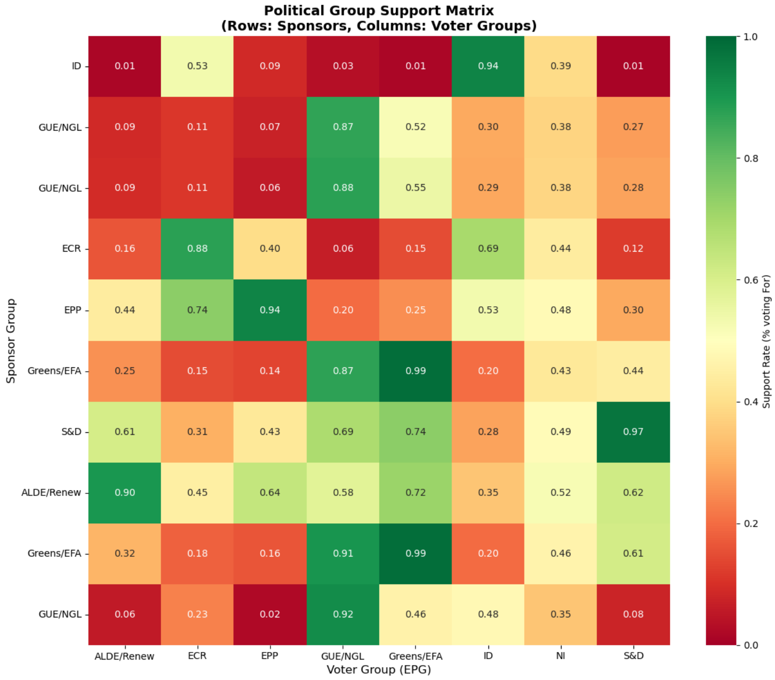

Figure 3: Support Matrix

Metodology:

- Step 1 → Data Integration
- Step 2 → Group Name Standardization: For each political group acting as sponsor: political groups have multiple naming variations across files (e.g., "EUL/NGL", "The Left", and "Confederal Group of the European United Left - Nordic Green Left" all refer to GUE/NGL). We created a mapping dictionary to standardize all group names to short, consistent labels.
- Step 3 → Vote Matching, for each political group acting as sponsor:
  - Identified all votes in the Voted_docs file where that group is listed as Author

- Matched these votes to the corresponding columns in the RCV file using position-based indexing
- Extracted the voting behavior of all MEPs on those specific votes

- Step 4 $\rightarrow$ Support Rate Calculation, for each (sponsor, voter) pair
  - Filtered votes to only those sponsored by the sponsor group
  - Extracted how MEPs from the voter group voted on those proposals
  - Applied the formula Support Rate $= \frac{(\#\text{ of For votes})}{(\#\text{ of For + Against + Abstain})}$

Key findings:

- **Very high in-group loyalty** across all groups (87-99%), with Greens/EFA (99%) and S&D (97%) showing the strongest discipline. This indicates that European political groups function as cohesive voting blocs with strong party discipline.
- **Three distinct voting blocks**
  - Left alliance (S&D , Greens/EFA , GUE/NGL): Mutual support ranging from 61-99%
  - Center-right coallition (EPP , ECR): Moderate cooperation (40-74%)
  - Liberal bridge (ALDE/Renew): Supports both left and right groups moderately (62-72%), acting as a swing voter
- **Political isolatio of the far-right** The ID group (Identity and Democracy) receives near-zero support from all other groups (1-9%), while maintaining high internal loyalty (94%)
- **Grand Coalition** EPP and S&D, traditionally the two largest groups, show asymmetric support (EPP$\rightarrow$S&D: 30%, S&D$\rightarrow$EPP: 43%). While they cooperate on major legislation, the moderate support levels confirm that ideological differences remain significant and coalition formation requires negotiation rather than automatic agreement.
- **Strong left-wing cooperation** The highest cross-group support rates appear among left-wing groups, particularly Greens/EFA supporting GUE/NGL proposals (91-99%), indicating that the left bloc votes together more cohesively than the right.

MDS showed that groups vote similarly while Support matrix shows who leads and who follows (e.g., Greens support S&D proposals 74%, but S&D supports Greens 61% - slightly asymmetric). This makes the support matrix analysis particularly valuable for understanding legislative effectiveness and coalition formation in the European Parliament rather than just having an overall ideological structure with MDS.

C Section

Question C.1

Considering a dataset matrix $Y \in R^{d \times n}$, its SVD $Y = U\Sigma V^T$ and the similarity matrix $S = Y^T Y$, we want to prove that $S = V\Sigma^2 V^T$

$$\begin{aligned} S = Y^T Y &= (U\Sigma V^T)^T (U\Sigma V^T) \\ &= V\Sigma^T U^T U \Sigma V^T \end{aligned}$$

Since U is known to be orthonormal from the SVD definition

$$U^T U = I$$

$$S = V\Sigma^T \Sigma V^T$$

Σ is a diagonal matrix with no-negative entries

$$\Sigma^T = \Sigma \quad \Sigma^T \Sigma = \Sigma^2$$

We finally obtain

$$S = V\Sigma^2 V^T$$

Question C.2

We want to prove that the two embeddings (MDS and PCA) differ only by an orthogonal transform such that $X_{PCA} = RX_{MDS}$ and that for Euclidean data MDS and PCA have the yield to the same embedding (up to rotation)
Theoretical concepts:

- MDS embedding
$$S = U\Lambda U^{-1} = U\Lambda U^T$$
$$X_{MDS} = I_{k \times n}\Lambda^{0.5} U^T = \Lambda_k^{0.5} U_k^T$$
Λ_k is the diagonal matrix of top k eigenvalues while U_k represents the first k columns of U (top k eigenvectors)

- PCA embedding
$$Y = U^{SVD}\Sigma V^T$$
$$X_{PCA} = (U_k^{SVD})^T Y$$
Knowing that $S = Y^T Y = V\Sigma^2 V^T$ and $S = U\Lambda U^T$
  - Eigenvectors of S are the right singular vectors V of Y
  - Eigenvalues of S are the square singular values $\lambda_i = \sigma_i^2$
$$U = V \quad \Lambda = \Sigma^2$$

We can rewrite both X_{MDS} and X_{PCA} to find a common definition:

- $X_{MDS} = \Sigma_k V_k^T$
  Replacing $U = V \quad \Lambda = \Sigma^2$ into its original definition $X_{MDS} = \Lambda_k^{0.5} U_k^T$
- $X_{PCA} = (U_k^{SVD})^T Y = (U_k^{SVD})^T (U_{SVD} \Sigma V^T) = \Sigma_k V_k^T$
  We have obtained the final form by using this expression $(U_k^{SVD})^T U^{SVD} = [I_k|0]$

We can compare the two expressions and see that:

$$X_{PCA} = \Sigma_k V_k^T \quad X_{MDS} = \Sigma_k V_k^T$$

$$X_{PCA} = I_k X_{MDS}$$

Where I_k is the identity matrix $R^{k \times k}$ which is orthogonal. We have proved that:

$$R = I_k$$

For Euclidean data (zero centered), classical MDS yields exactly the same embedding as PCA (they differ by an orthogonal transformation with $R = I_k$ which is the identity rotation)
In simpler words: MDS works on distances while PCA works on variances. They only align when the data is centered.

Question C.3

We want to show that the variance of the dataset $Var(Y) = \sum_{y \in Y} ||y - \bar{y}||_2^2$ can be expressed as $Var(Y) = \sum_{i=1}^{d} \sigma_i^2$.
We know that the data is zero-centered ($\bar{y} = 0$)

$$\begin{aligned}
Var(y) &= \sum_{y \in Y} ||y||_2^2 \\
&= \sum_{i=1}^{n} ||y_i||_2^2 \\
&= \sum_{i=1}^{n} \sum_{j=1}^{d} y_{ij}^2 \\
&= ||Y||_F^2
\end{aligned}$$

$||Y||_F^2$ is the Frobenius Norm.
We want to show that:

$$||Y|||_F^2 = \sum_{i=1}^{d} \sigma_i^2$$

We can start noticing that the Frobenius norm can be expressed using the trace

$$
\begin{aligned}
\text{Trace}(Y^TY) &= \sum_{j=1}^{n}(Y^TY)_{jj} \\
&= \sum_{j=1}^{n}\sum_{i=1}^{d} Y^T_{ji}Y_{ij} \\
&= \sum_{j=1}^{n}\sum_{i=1}^{d} Y_{ij}Y_{ij} \\
&= \sum_{j=1}^{n}\sum_{i=1}^{d} Y_{ij}^2 \\
&= ||Y||_F^2
\end{aligned}
$$

We can prove that multiplying with an orthonormal matrix ($U^TU = I$ or $V^TV = I$) does not change the Frobenius norm:

$$
\begin{aligned}
\|UY\|_F^2 &= \text{tr}((UY)^T(UY)) \\
&= \text{tr}(Y^TU^TUY) \\
&= \text{tr}(Y^TY) \\
&= \|Y\|_F^2
\end{aligned}
\qquad\qquad
\begin{aligned}
\|YV^T\|_F^2 &= \text{tr}((YV^T)^T(YV^T)) \\
&= \text{tr}(VY^TYV^T) \\
&= \text{tr}(Y^TYV^TV) \\
&= \text{tr}(Y^TY) \\
&= \|Y\|_F^2
\end{aligned}
$$

Knowing these properties and the SVD $Y = U\Sigma V^T$, we can finally show that:

$$
\begin{aligned}
||Y||_F^2 &= ||U\Sigma V^T||_F^2 \\
&= ||\Sigma V^T||_F^2 \\
&= ||\Sigma||_F^2 \\
&= \sum_{i=1}^{n}\sigma_i^2
\end{aligned}
$$

Question C.4

We want to show that the variance of the projected data X is given by $Var(X) = \sum_{i=1}^{k}\sigma_i^2$

$$X = W^TY$$

$$W \in R^{d,k} \quad X \in R^{k,n} \quad Y \in R^{d,n}$$

$$
\begin{aligned}
Var(X) &= \sum_{i=1}^{n} ||x_i - \bar{x_i}||_2^2 \\
&= \sum_{i=1}^{n} ||x_i||_2^2
\end{aligned}
$$

Since Y is zero centered ($\bar{y} = 0$),the projected data is also zero centered:

$$
\begin{aligned}
\bar{x} &= \frac{1}{n}\sum_{j=1}^{n} x_j \\
&= \frac{1}{n}\sum_{j=1}^{n} W^T y_j \\
&= W^T \bar{y} \\
&= 0
\end{aligned}
$$

The proof is very similar to the previous exercise , we will exploit the trace its cyclic property.

$$
\begin{aligned}
Var(X) &= ||X||_F^2 \\
&= ||W^T Y||_F^2 \\
&= \text{Trace}((W^T Y)^T (W^T Y)) \\
&= \text{Trace}(Y^T W W^T Y) \\
&= \text{Trace}(W W^T Y Y^T) \\
&= \text{Trace}(W W^T U \Sigma V^T V \Sigma^T U^T) \\
&= \text{Trace}(W W^T U \Sigma^2 U^T)
\end{aligned}
$$

We know that $W = U_k$ from the slide of the Module 7 (PCA theory)

$$
\begin{aligned}
Var(X) &= \text{Trace}(U_k U_k^T U \Sigma^2 U^T) \\
&= \text{Trace}(U_k^T U \Sigma^2 U^T U_k)
\end{aligned}
$$

We know that:

$$
U_k^T U = (I_k|0) \quad U^T U_k = \begin{pmatrix} I_k \\ 0 \end{pmatrix}
$$

It becomes:

$$
\begin{aligned}
Var(X) &= \text{Trace}\left((I_k|0)\Sigma^2 \begin{pmatrix} I_k \\ 0 \end{pmatrix}\right) \\
&= \sum_{i=1}^{k} \sigma_i^2
\end{aligned}
$$

Question C.5

We consider the residual points $Z = z_1, ..., z_n$ where $z_i = y_i - WW^Ty_i$ and we want to show that its variance is given by $Var(Z) = \sum_{i=k+1}^{d} \sigma_i^2$
We start from the definition of variance

$$\begin{aligned} Var(Z) &= \sum_{i=1}^{n} ||z_i - \bar{z}||_2^2 \\ &= \sum_{i=1}^{n} ||z_i||_2^2 \\ &= ||Y - WW^TY||_F^2 \end{aligned}$$

The residual points are zero centered, below we have the proof for that.

$$\begin{aligned} \bar{z} &= \frac{1}{n}\sum_{i=1}^{n} y_i - WW^Ty_i \\ &= \frac{1}{n}\sum_{i=1}^{n} y_i - WW^Ty_i \\ &= \frac{1}{n}\sum_{i=1}^{n} y_i - \frac{1}{n}\sum_{i=1}^{n} WW^Ty_i \\ &= \bar{y} - WW^T\bar{y} \\ &= 0 \end{aligned}$$

We will use the trace, its cyclic and linearty property for the proof, it is also very important to remmber that $W^TW = I_k$

$$\begin{aligned} Var(Z) &= ||Y - WW^TY||_F^2 \\ &= \text{Trace}((Y - WW^TY)^T(Y - WW^TY)) \\ &= \text{Trace}((Y^T - Y^TWW^T)(Y - WW^TY)) \\ &= \text{Trace}(Y^TY - Y^TWW^TY - YY^TWW^T + Y^TWW^TWW^TY) \\ &= \text{Trace}(Y^TY - 2Y^TWW^TY + Y^TWW^TWW^TY) \\ &= \text{Trace}(Y^TY - 2Y^TWW^TY + Y^TWW^TY) \\ &= \text{Trace}(Y^TY - Y^TWW^TY) \\ &= \text{Trace}(Y^TY) - \text{Trace}(Y^TWW^TY) \\ &= \text{Trace}(Y^TY) - \text{Trace}(WW^TYY^T) \end{aligned}$$

Both of them have already been seen in the previous exercises, in particular:

- $\text{Trace}(Y^TY)$ has been discussed in the Question C.3 where we proved that

$$\text{Trace}(Y^TY) = \sum_{i=1}^{d} \sigma_i^2$$

- Trace(WW^TYY^T) has been discussed in the Question C.4 where we proved that

$$\text{Trace}(WW^TYY^T) = \sum_{i=1}^{k} \sigma_i^2$$

The final expression becomes:

$$\begin{aligned} Var(Z) &= \text{Trace}(Y^TY) - \text{Trace}(WW^TYY^T) \\ &= \sum_{i=1}^{d} \sigma_i^2 - \sum_{i=1}^{k} \sigma_i^2 \\ &= \sum_{i=k+1}^{d} \sigma_i^2 \end{aligned}$$

We can conclude our discussion saying that:

$$Var(Y) = Var(X) + Var(Z)$$

$$Var(Y) = \sum_{i=1}^{k} \sigma_i^2 + \sum_{i=k+1}^{d} \sigma_i^2$$

Where Var(Y) is the total variance and it can be computed by taking into account the variance explained by PCA (Var(X)) and the residual variance (Var(Z))

B Section

Question B.1

D is the degree matrix, a diagonal matrix where:

- $D_{ii} = \sum_j A_{ij}$ which is the degree of node i (the number of edges connected to it)
- $D_{ij} = 0$ for all $i \neq j$ (off-diagonal entries are zero)

P is the transition probability matrix for a random walk on the graph:

$$P_{ij} = \frac{A_{ij}}{D_{ii}} = \frac{A_{ij}}{\text{degree of node i}}$$

P_{ij} is the probability of moving from node i to node j in one step if you randomly choose one of node i's neighbors.

The similarity measure S is defined using the variable k:

$$S_{ij} = \sum_{k=1}^{\infty} \alpha^k P_{ij}^k$$

k is the number of steps (or path length) in a random walk:

- k=1: one-step paths (direct neighbors)
- k=2: two-step paths (neighbors of neighbors)
- k=3: three-step paths, and so on...

Let's go deeper into the definition of the similarity measure:

- P_{ij}^k is the probability of reaching node j from node i in exactly k steps via a random walk. A large value indicates many k-step paths exist between i and j.
- α^k is the decay factor: it gives exponentially decreasing weight to longer paths, ensuring direct connections (small k) contribute more than distant connections (large k)
  - The decay parameter balances local vs. global structure: smaller α emphasizes direct connections, while larger α considers longer-range connections.
  - The decay parameter ensures we don't count very long, indirect paths too heavily (nearby structure matters more than distant connections).
  - The condition on the decay parameter ($0 < \alpha < 1$) allows the infinite sum to converge to a finite value
- $\sum_{k=1}^{\infty}$ sums the contributions from all path lengths, combining information from direct connections, common neighbors, and broader network structure.

S captures the idea that two nodes are similar if they are connected through many paths of various lengths.

Question B.2

For all $i, j \in V$, we want to prove that

- $P_{ij} \leq 1$

$$P = D^{-1}A$$
$$P_{ij} = \frac{A_{ij}}{D_{ii}} = \frac{A_{ij}}{\sum_j A_{ij}}$$

  Because A_{ij} is either 0 or 1, and D_{ii} is the sum of all entries in row i, it follows that $A_{ij} \leq D_{ii}$. Consequently, the ratio is a binary value divided by a positive integer, which implies $0 \leq P_{ij} \leq 1$.

- $S_{ij} < +\infty$

$$S_{ij} = \sum_{k=1}^{\infty} \alpha^k P_{ij}^k$$
$$S = \sum_{k=1}^{\infty} \alpha^k P^k = \sum_{k=0}^{\infty} (\alpha P)^k - I$$

It is a geometric series which will converge to a finite value if $-1 < \alpha P < 1$: we know that $P_{ij} < 1$ and by definition of the problem we can assume $0 < \alpha < 1$.

$$S = (I - \alpha P)^{-1} - I$$

Since the result is a well-defined finite matrix, every entry S_{ij} must be finite ($S_{ij} < +\infty$).

Question B.3

We want to show that S can be computed efficiently using a matrix inversion operation.

$$\begin{aligned} S &= \sum_{k=1}^{\infty} \alpha^k P^k \\ &= \sum_{k=0}^{\infty} (\alpha P)^k - I \\ &= (I - \alpha P)^{-1} - I \end{aligned}$$

Complexity:

- Compute $I - \alpha P$ is $O(n^2)$
- Invert $I - \alpha P$ is $O(n^3)$

Total complexity is then $O(n^3)$ with $n = |V|$

A Section

Multidimensional Scaling (MDS) is a dimensionality reduction technique that aims to embed high-dimensional data into a lower-dimensional space while preserving pairwise distances. Given a distance matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ containing all pairwise distances d_{ij} between n data points, the goal is to find a k-dimensional representation $\mathbf{x}_i \in \mathbb{R}^k$ for each point i, where typically $k \ll n$. These vectors can be arranged into a configuration matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{k \times n}$.

The quality of the embedding is measured by the stress function:

$$E(\mathbf{X}) = \sum_{i<j} w_{ij} \left(d_{ij} - \|\mathbf{x}_i - \mathbf{x}_j\| \right)^2, \tag{1}$$

where w_{ij} are weights that can either be specified as input or computed from the distances themselves. This objective quantifies the discrepancy between the original distances and the Euclidean distances in the embedded space.

Since the stress function does not admit a closed-form solution, numerical optimization methods such as gradient descent are employed to find the configuration $\mathbf{X}$ that minimizes $E(\mathbf{X})$.

Question A.1

We want to show that the gradient of E wrt a point x_i is

$$\nabla_{x_i} E = 2 \sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{||x_i - x_j||} \right) (x_i - x_j)$$

We start from the definition of E:

$$E(X) = \sum_{i<j} w_{ij} \left(d_{ij} - \|x_i - x_j\| \right)^2$$

We apply the gradient wrt x_i and we used the linearity and chain rule property of it.

$$\begin{aligned}
\nabla_{x_i} E(x) &= \nabla_{x_i} \left(\sum_{i<j} w_{ij} \left(d_{ij} - \|x_i - x_j\| \right)^2 \right) \\
&= \nabla_{x_i} \left(\frac{1}{2} \sum_{k=1}^{n} \sum_{j \neq k} w_{kj} \left(d_{kj} - \|x_k - x_j\| \right)^2 \right) w
\end{aligned}$$

Here, the factor of $\frac{1}{2}$ accounts for counting each pair twice: once as (k,j) and once as (j,k). When we take $\nabla_{\mathbf{x}_i}$ we need to find which terms in the double sum depend on x_i, it appears in:

- The outer sum when k=i: terms of the form $\|\mathbf{x}_i - \mathbf{x}_j\|$ for all $j \neq i$
- The inner sum when j=i: terms of the form $\|\mathbf{x}_k - \mathbf{x}_i\|$ for all $k \neq i$

The expression becomes

$$\nabla_{\mathbf{x}_i} E = \frac{1}{2} \left[\sum_{j \neq i} \nabla_{\mathbf{x}_i} \left(w_{ij} (d_{ij} - \|\mathbf{x}_i - \mathbf{x}_j\|)^2 \right) + \sum_{k \neq i} \nabla_{\mathbf{x}_i} \left(w_{ki} (d_{ki} - \|\mathbf{x}_k - \mathbf{x}_i\|)^2 \right) \right]$$

We can observe that the distances (d) and the weights are (w) are symmetric and both the sums have the same exact form, it becomes:

$$\begin{aligned}
\nabla_{x_i} E(x) &= \sum_{i \neq j} w_{ij} \nabla_{x_i} \left((d_{ij} - \|x_i - x_j\|)^2 \right) \\
&= \sum_{i \neq j} 2 w_{ij} \left(d_{ij} - \|x_i - x_j\| \right) \nabla_{x_i} \left(-||x_i - x_j|| \right) \\
&= \sum_{i \neq j} 2 w_{ij} \left(d_{ij} - \|x_i - x_j\| \right) \left(- \frac{x_i - x_j}{||x_i - x_j||} \right) \\
&= 2 \sum_{i \neq j} w_{ij} \left(1 - \frac{d_{ij}}{||x_i - x_j||} \right) (x_i - x_j)
\end{aligned}$$

Question A.2

Considering a matrix $A \in R^{k \times k}$ be an orthonormal matrix, b $\in R^k$ and $1_n \in R^n$ a vector of all 1's. We want to show that $E(AX + b1_n^T) = E(X)$.

$$\begin{aligned}
E(AX + b1_n^T) &= \sum_{i<j} w_{ij} \left(d_{ij} - \|Ax_i + b1_n^T - Ax_j - b1_n^T\|\right)^2 \\
&= \sum_{i<j} w_{ij} \left(d_{ij} - \|Ax_i - Ax_j\|\right)^2 \\
&= \sum_{i<j} w_{ij} \left(d_{ij} - \|A(x_i - Ax_j)\|\right)^2 \\
&= \sum_{i<j} w_{ij} \left(d_{ij} - \|x_i - x_j\|\right)^2 \\
&= E(X)
\end{aligned}$$

Because the stress $E(X)$ remains identical regardless of how we rotate (A) or shift (b) the entire configuration, the local optima are not unique. Any optimal solution can be transformed into another optimal solution with the same stress value.
We used this property before $||Ax|| = ||x||$ with A orthonormal. We want to prove it in order to have a better understanding:

$$\begin{aligned}
||Ax||^2 &= (Ax)^T(Ax) \\
&= x^T A^T Ax \\
&= x^T x \\
&= ||x||^2
\end{aligned}$$

Taking square roots we have proved that $||Ax|| = ||x||$

Question A.3

We want to prove that $OPT(k_2) \leq OPT(k_1)$ where $k_1, k_2 \in N$ with $0 < k_1 < k_2$ and the optimal stress in dimension k is defined as $OPT(k) = \inf_{X \in R^{k \times n}} E(X)$
Let $\mathbf{X}^* \in \mathbb{R}^{k_1 \times n}$ be an optimal solution for dimension k_1. This means that $\mathbf{X}^*$ minimizes the stress function E over all possible configurations in $\mathbb{R}^{k_1 \times n}$. Therefore:

$$E(\mathbf{X}^*) = \mathrm{OPT}(k_1)$$

We denote the columns of $\mathbf{X}^*$ as $\mathbf{x}_1^*, \mathbf{x}_2^*, \ldots, \mathbf{x}_n^*$, where each $\mathbf{x}_i^* \in \mathbb{R}^{k_1}$ represents the coordinates of point i in the k_1-dimensional embedding.

Now we construct a configuration $\tilde{\mathbf{X}} \in \mathbb{R}^{k_2 \times n}$ by embedding $\mathbf{X}^*$ into the higher-dimensional space $\mathbb{R}^{k_2}$. We do this by appending $(k_2 - k_1)$ zero coordinates to each point:

$$\tilde{\mathbf{X}} = \begin{bmatrix} \mathbf{X}^* \\ \mathbf{0} \end{bmatrix} \in \mathbb{R}^{k_2 \times n}$$

where $\mathbf{0} \in \mathbb{R}^{(k_2-k_1)\times n}$ is a matrix containing only zeros.

More explicitly, for each point $i = 1, 2, \ldots, n$, we define:

$$\tilde{\mathbf{x}}_i = \begin{bmatrix} \mathbf{x}_i^* \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} x_{i,1}^* \\ x_{i,2}^* \\ \vdots \\ x_{i,k_1}^* \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{k_2}$$

where the last $(k_2 - k_1)$ components are all zero.

Now we compute the Euclidean distance between any two points $\tilde{\mathbf{x}}_i$ and $\tilde{\mathbf{x}}_j$ in this higher-dimensional space:

$$\begin{aligned}
\|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|^2 &= \left\| \begin{bmatrix} \mathbf{x}_i^* \\ \mathbf{0} \end{bmatrix} - \begin{bmatrix} \mathbf{x}_j^* \\ \mathbf{0} \end{bmatrix} \right\|^2 \\
&= \left\| \begin{bmatrix} \mathbf{x}_i^* - \mathbf{x}_j^* \\ \mathbf{0} \end{bmatrix} \right\|^2 \\
&= \sum_{\ell=1}^{k_1} (x_{i,\ell}^* - x_{j,\ell}^*)^2 + \sum_{\ell=k_1+1}^{k_2} 0^2 \\
&= \sum_{\ell=1}^{k_1} (x_{i,\ell}^* - x_{j,\ell}^*)^2 \\
&= \|\mathbf{x}_i^* - \mathbf{x}_j^*\|^2
\end{aligned}$$

Taking square roots on both sides:

$$\|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\| = \|\mathbf{x}_i^* - \mathbf{x}_j^*\|$$

This shows that the Euclidean distances between all pairs of points are preserved under this embedding. The additional zero coordinates do not contribute to the distance calculation.

Now we compute the stress value for the configuration $\tilde{\mathbf{X}}$ in the k_2-dimensional space. Using the definition of the stress function from Equation (2):

$$E(\tilde{\mathbf{X}}) = \sum_{i<j} w_{ij} \left(d_{ij} - \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|\right)^2$$

Substituting the distance preservation property we just proved:

$$\begin{aligned}
E(\tilde{\mathbf{X}}) &= \sum_{i<j} w_{ij} \left(d_{ij} - \|\mathbf{x}_i^* - \mathbf{x}_j^*\|\right)^2 \\
&= E(\mathbf{X}^*)
\end{aligned}$$

Since $\mathbf{X}^*$ was the optimal configuration in dimension k_1, we have:

$$E(\tilde{\mathbf{X}}) = E(\mathbf{X}^*) = \mathrm{OPT}(k_1)$$

Now, observe that $\tilde{\mathbf{X}} \in \mathbb{R}^{k_2 \times n}$ is a feasible configuration in the k_2-dimensional space. By definition, $\mathrm{OPT}(k_2)$ is the minimum value of the stress function over all possible configurations in $\mathbb{R}^{k_2 \times n}$:

$$\mathrm{OPT}(k_2) = \min_{\mathbf{Y} \in \mathbb{R}^{k_2 \times n}} E(\mathbf{Y})$$

Since $\tilde{\mathbf{X}}$ is one particular configuration in $\mathbb{R}^{k_2 \times n}$, and the minimum is taken over all such configurations, we must have:

$$\mathrm{OPT}(k_2) \leq E(\tilde{\mathbf{X}})$$

Combining this with our earlier result:

$$\mathrm{OPT}(k_2) \leq E(\tilde{\mathbf{X}}) = E(\mathbf{X}^*) = \mathrm{OPT}(k_1)$$

Therefore:

$$\mathrm{OPT}(k_2) \leq \mathrm{OPT}(k_1)$$

This proves that the minimum achievable stress in dimension k_2 is at most the minimum achievable stress in dimension k_1. In other words, as we increase the embedding dimension, the optimal stress value can only decrease or remain the same, but it cannot increase.

The intuition behind this result is that higher-dimensional spaces provide more flexibility and degrees of freedom for positioning the points. Any configuration that exists in a lower-dimensional space can be embedded into a higher-dimensional space (by adding zero coordinates) without changing the pairwise distances. Therefore, the set of achievable distance configurations in higher dimensions is at least as large as in lower dimensions, which means we can achieve at least as good an approximation to the target distances d_{ij}.

Question A.4

We want to explain why a local optima of the stress function satisfy

$$\sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{\|\mathbf{x}_i - \mathbf{x}_j\|}\right) (\mathbf{x}_i - \mathbf{x}_j) = \mathbf{0}$$

We found the gradient of the stress function in the question A.1, which is:

$$\nabla_{x_i} E = 2 \sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{||x_i - x_j||}\right) (x_i - x_j)$$

A local optima of a differentiable function must satisfy the first-order optimality condition: the gradient must be zero $\nabla_{x_i} E = 0$.

$$\nabla_{x_i} E = 2 \sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{||x_i - x_j||}\right) (x_i - x_j) = 0$$

We divide by the constant 2 , in order to get the desired expression

$$\sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{||x_i - x_j||}\right) (x_i - x_j) = 0$$

This represents a balance of forces: each point experiences forces from all other points.

Question A.5

The local optimum condition from Question A.4 can be written as:

$$\sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|}\right) (x_i - x_j) = 0$$

This represents a **balance of forces** acting on each point x_i. Each neighboring point x_j exerts a force on x_i along the direction $(x_i - x_j)$ with magnitude proportional to $w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|}\right)$.

Case (i): $\|x_i - x_j\| > d_{ij}$ (Points are too far apart)

When the embedded distance exceeds the target distance:

$$1 - \frac{d_{ij}}{\|x_i - x_j\|} > 0$$

The force is in the direction $(x_i - x_j)$, which points *away* from x_j. However, since this represents a spring-like attraction, the positive coefficient means point x_j exerts an attractive force on x_i, pulling it closer to reduce the excessive distance.

Interpretation: Points that are too far apart attract each other to decrease their separation.

Case (ii): $\|x_i - x_j\| < d_{ij}$ (Points are too close)

When the embedded distance is less than the target distance:

$$1 - \frac{d_{ij}}{\|x_i - x_j\|} < 0$$

The force has a negative coefficient, reversing the direction. Point x_j now exerts a repulsive force on x_i, pushing it away to increase their separation.

Interpretation: Points that are too close together repel each other to increase their distance.

Equilibrium

At a local optimum, the sum of all attractive and repulsive forces on each point equals zero. This creates a configuration where:

- Pairs with $\|x_i - x_j\| > d_{ij}$ contribute attractive forces
- Pairs with $\|x_i - x_j\| < d_{ij}$ contribute repulsive forces
- The weighted sum of these forces balances out at each point

This mechanical analogy is similar to a spring-mass system where springs have rest lengths d_{ij} and stiffness w_{ij}, and the system reaches equilibrium when all forces balance.